Presentation Speaker Notes

CSCE 5300 | Project Group 7

Slide 1: Parallel K-Means Clustering (30 seconds)

Introduce Group 7, the research topic, and the goal of evaluating parallel clustering performance.

Slide 2: Research Motivation (45 seconds)

Explain why clustering matters in big data analytics and why execution time becomes important as datasets grow.

Slide 3: Research Foundation (50 seconds)

Identify the approved paper and distinguish its proposed method from the implementation developed for this course project.

Slide 4: Traditional K-Means (55 seconds)

Walk through the iterative K-Means procedure. Explain that lower SSE means observations are closer to their assigned centers, but comparisons must use the same dataset and conditions.

Slide 5: Parallel K-Means Strategy (60 seconds)

Explain the partition-level initialization and subsequent global refinement. Avoid claiming that this pipeline is identical to every step of the published paper.

Slide 6: Implementation Architecture (55 seconds)

Describe the implemented software components and how Spark is used. Clarify that the tested local configurations run on one physical computer.

Slide 7: Experimental Design (60 seconds)

Explain the controlled variables and the repeated experimental design. local[1], local[2], and local[4] represent local execution parallelism, not three separate cluster machines.

Slide 8: Execution Time Comparison (60 seconds)

Discuss the runtime figure and the reduction in measured execution time as local parallelism increases.

Slide 9: Measured Speedup (60 seconds)

Explain the speedup calculation and why overhead, coordination, and hardware constraints can prevent perfect linear scaling.

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Slide 10: Clustering Quality (60 seconds)

Explain SSE and why the consistency across worker settings matters. Do not infer that the larger dataset has inherently worse clustering solely from its larger total SSE.

Slide 11: Traditional Baseline Comparison (55 seconds)

Review the baseline figure against the final report before presenting. Do not make numerical baseline claims that have not been verified from the original baseline data.

Slide 12: Research Comparison and Limitations (55 seconds)

Make a careful qualitative comparison with the paper. Check the report's paper-comparison section before adding any quantitative claims about published results.

Slide 13: Live Implementation Demonstration (90 seconds)

Run a short, rehearsed demonstration. Confirm the exact command and expected output before presentation day.

Slide 14: Conclusions (50 seconds)

Summarize the findings without claiming that local-mode experiments prove general distributed scalability.

Slide 15: References and Questions (30 seconds)

Before submission, replace the abbreviated paper reference with the complete verified IEEE citation and add any references needed for slide-specific claims.